

∴ Linear Partial Differential Equations of Order one.

Forming PDE by elimination of Arbitrary constants/Functions

→ An equation involving partial differential coefficients of a function of two or more variables is known as a partial differential equation. We use the following notations for partial differential equations.

$$\boxed{p = \frac{\partial z}{\partial x} \mid q = \frac{\partial z}{\partial y} \mid r = \frac{\partial^2 z}{\partial x \partial y} \mid t = \frac{\partial^2 z}{\partial y^2}}$$

1) $z = (ax + \frac{y}{a} + b)^2$

2) $z = Ae^{pt} \cdot \sin(px)$

3) $z = a(x+y) + b(x-y) + abt$

4) $z = axe^y + \frac{1}{2}a^2e^y + b$

→ Objective: removing arbitrary constants

5) $z = y^2 + 2f(\frac{x}{y} + \ln y)$

6) $z = f(x+iy) + F(x-iy), i^2 = -1$

7) $z = f(x-at) + F(x+at)$

8) $z = f(xy) + g(xy)$

→ Objective: removing arbitrary functions

→ arbitrary: Any value/function/constants that are not fixed.

→ We differentiate because, differentiating is the easiest way to make the constants/functions appear directly.

$$1) z = (ax + \frac{y}{a} + b)^2 \quad | \text{need to remove a \& b}$$

b) Differentiating with respect to x & y

$$\therefore \frac{\partial z}{\partial x} = 2(ax + \frac{y}{a} + b) \cdot \frac{\partial}{\partial x} (ax + \frac{y}{a} + b)$$

$$\& \quad = 2(ax + \frac{y}{a} + b) \cdot a$$

$$\therefore \frac{\partial z}{\partial y} = 2(ax + \frac{y}{a} + b) \cdot \frac{\partial}{\partial y} (ax + \frac{y}{a} + b)$$

$$= 2(ax + \frac{y}{a} + b) \cdot \frac{1}{a}$$

Again,

$$\frac{\partial^2 z}{\partial x^2} = 2a \cdot a = 2a^2 \quad \& \quad \frac{\partial^2 z}{\partial y^2} = \frac{2}{a} \cdot \frac{1}{a} = \frac{2}{a^2}$$

$$\Rightarrow a^2 \cdot \frac{2}{a^2} = 2$$

putting value in eqn ①

$$\therefore \frac{\partial^2 z}{\partial x^2} = 2 \cdot \left(\frac{2}{a^2} \right) = \frac{4}{a^2}$$

$$\Rightarrow \left(\frac{\partial^2 z}{\partial x^2} \right) \left(\frac{\partial^2 z}{\partial y^2} \right) = 4 \quad (M)$$

② $z = Ae^{pt} \cdot \sin px$ | need to remove A & p |

Differentiating with respect to x & t .

$\therefore \frac{\partial z}{\partial x} = Ae^{pt} \cdot p \cos px$ & $\frac{\partial z}{\partial t} = Ape^{pt} \cdot \sin px$

Again,

$\frac{\partial^2 z}{\partial x^2} = -Ae^{pt} \cdot p^2 \sin px$ & $\frac{\partial^2 z}{\partial t^2} = Ap^2e^{pt} \cdot \sin px$

$z = -Ap^2e^{pt} \cdot \sin px$ $\rightarrow$ ①

$\therefore \frac{\partial^2 z}{\partial x^2} = -\frac{\partial^2 z}{\partial t^2}$

$\Rightarrow \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0$

③ $z = a(x+y) + b(x-y) + abt + c$ | need to remove a & b & c

$\Rightarrow z = ax + ay + bx - by + abt + c$

$\therefore$ Differentiating with respect to x, y & t

$\Rightarrow \frac{\partial z}{\partial x} = a+b$ & $\frac{\partial z}{\partial y} = a-b$ & $\frac{\partial z}{\partial t} = ab$

We know,

$ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$

$\Rightarrow \frac{\partial z}{\partial t} = \left(\frac{\partial z / \partial x}{2}\right)^2 - \left(\frac{\partial z / \partial y}{2}\right)^2$

$\Rightarrow 4 \frac{\partial z}{\partial t} = \left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2$ (A)

(4) $z = axe^y + \frac{1}{2}a^2e^y + b$ I need to remove a & b

∴ Differentiating with respect to x & y

⇒ $\boxed{\frac{\partial z}{\partial x} = ae^y}$ & $\frac{\partial z}{\partial y} = axe^y + \frac{1}{2}a^2e^y$

⇒ $\boxed{\frac{\partial z}{\partial x} \cdot \frac{1}{e^y} = a}$ $\frac{\partial z}{\partial y} = ae^y \left(x + \frac{1}{2}a \right)$ → (1)

∴ putting values in eqn (1)

⇒ ~~$\frac{\partial z}{\partial y}$~~

∴ $\frac{\partial z}{\partial y} = \frac{\partial z}{\partial x} \left(x + \frac{1}{2} \cdot \frac{1}{e^y} \cdot \frac{\partial z}{\partial x} \right)$

⇒ $\frac{\partial z}{\partial y} = x \frac{\partial z}{\partial x} + \frac{1}{2e^y} \left(\frac{\partial z}{\partial x} \right)^2$ (Ans)

⇒ $q = px + \frac{p^2}{2e^y}$ (Ans)

∴ d & a given at hand | $c + tda + (y-x)d + (y+x)a = \frac{1}{2}$ (2)

$cx + tda + yd - xd + ya + xa = \frac{1}{2}$

∴ Differentiating with respect to x & y

$\frac{da}{dx} = \frac{1}{2} \quad \& \quad d - a = \frac{1}{2} \quad \& \quad d + a = \frac{1}{2}$

$\frac{1}{2} \left(\frac{d-a}{2} \right) - \frac{1}{2} \left(\frac{d+a}{2} \right) = da$

$\frac{1}{2} \left(\frac{1-1}{2} \right) - \frac{1}{2} \left(\frac{1+1}{2} \right) = \frac{1}{2} da$

5) $z = y^2 + 2f\left(\frac{1}{x} + \ln y\right)$ | Need to remove f |

$\therefore$ Differentiating with respect to x & y

$$\begin{aligned} \Rightarrow \frac{\partial z}{\partial x} &= 2f'\left(\frac{1}{x} + \ln y\right) \cdot \frac{\partial}{\partial x}\left(\frac{1}{x} + \ln y\right) & \& \quad \frac{\partial z}{\partial y} = 2y + 2f'\left(\frac{1}{x} + \ln y\right) \cdot \frac{\partial}{\partial y}\left(\frac{1}{x} + \ln y\right) \\ &= 2f'\left(\frac{1}{x} + \ln y\right) \cdot (x^{-2}) & & \quad = 2y + 2f'\left(\frac{1}{x} + \ln y\right) \cdot \left(\frac{1}{y}\right) \quad \text{--- ①} \end{aligned}$$

$$\Rightarrow -x^2 \cdot \frac{\partial z}{\partial x} = 2f'\left(\frac{1}{x} + \ln y\right)$$

putting in eqn ①

$$\therefore \frac{\partial z}{\partial y} = 2y \cdot \left(-x^2 \cdot \frac{\partial z}{\partial x}\right) \cdot \frac{1}{y}$$

$$\Rightarrow \frac{\partial z}{\partial y} = 2y \cdot \left(-\frac{x^2}{y} \cdot \frac{\partial z}{\partial x}\right)$$

$$\Rightarrow q = 2y - \frac{px^2}{y} \quad (A)$$

6) $z = f(x+iy) + F(x-iy)$ | Need to remove f & F , $[i^2 = -1]$

Differentiating with respect to x & y

$$\Rightarrow \frac{\partial z}{\partial x} = f'(x+iy) \cdot 1 + F'(x-iy) \cdot 1 \quad \& \quad \frac{\partial z}{\partial y} = f'(x+iy) \cdot i + F'(x-iy) \cdot (-i)$$

Again

$$\begin{aligned} \Rightarrow \frac{\partial^2 z}{\partial x^2} &= f''(x+iy) \cdot 1 + F''(x-iy) \cdot 1 & \& \quad \frac{\partial^2 z}{\partial y^2} = f''(x+iy) \cdot i \cdot i + F''(x-iy) \cdot (-i) \cdot (-i) \\ &= f''(x+iy) + F''(x-iy) & & \quad = f''(x+iy) - F''(x-iy) \quad \text{--- ①} \end{aligned}$$

putting value in eqn ①

$$\Rightarrow \frac{\partial^2 z}{\partial y^2} = f''(x+iy) - F''(x-iy) \quad \text{--- ①}$$

$[i^2 = -1]$ given

$$\therefore \frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial x^2}$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = 0 \Rightarrow p + q = 0 \quad (A)$$

$$7) z = f(x-at) + F(x+at)$$

Differentiate with respect to x & t

$$\therefore \frac{\partial z}{\partial x} = f'(x-at) \cdot 1 + F'(x+at) \cdot 1 \quad \& \quad \frac{\partial z}{\partial t} = f'(x-at)(-a) + F'(x+at)(a)$$

Again,

$$\therefore \frac{\partial^2 z}{\partial x^2} = f''(x-at) \cdot 1 + F''(x+at) \cdot 1 \quad \& \quad \frac{\partial^2 z}{\partial t^2} = f''(x-at)(-a \cdot -a) + F''(x+at)(a \cdot a)$$

$\rightarrow \textcircled{1}$

$$\Rightarrow a^2 \frac{\partial^2 z}{\partial t^2} = f''(x-at) + F''(x+at)$$

putting value in eqⁿ $\textcircled{1}$

$$\therefore \frac{\partial^2 z}{\partial x^2} = \frac{1}{a^2} \frac{\partial^2 z}{\partial t^2}$$

$$\Rightarrow a^2 \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial t^2}$$

$$\textcircled{8} z = f(xy) + g(xy) \quad | \text{Need to remove } f \& g |$$

$\therefore$ Differentiating with respect to x & y

$$\Rightarrow \frac{\partial z}{\partial x} = f(xy) \cdot y + g(xy) \cdot y \quad \& \quad \frac{\partial z}{\partial y} = f(xy) \cdot x + g(xy) \cdot x$$

$$\Rightarrow \frac{\partial z}{\partial x} = y \{ f(xy) + g(xy) \}$$

$$\Rightarrow \frac{\partial z}{\partial y} = x \{ f(xy) + g(xy) \}$$

$$\Rightarrow \frac{1}{y} \cdot \frac{\partial z}{\partial x} = \text{same}$$

$$\Rightarrow \frac{1}{x} \cdot \frac{\partial z}{\partial y} = \text{same}$$

$\rightarrow \textcircled{1}$

$\rightarrow \textcircled{1}$

$$\Rightarrow \frac{1}{y} \frac{\partial z}{\partial x} = \frac{1}{x} \frac{\partial z}{\partial y}$$

$$\Rightarrow x \frac{\partial z}{\partial x} = y \frac{\partial z}{\partial y} \quad (A)$$